



CYBER SECURITY AWARENESS

01 Oct 24



AIM



To foster cyber security awareness among all members of BAF to ensure security and safety in the cyberspace



SEQUENCE

- Cyberspace & Cyber Security
- Different Types of Common Cyber Attacks
- Effects of Cyber Attacks
- Cyber Safety Measures through Awareness

CYBERSPACE

Definition

Cyberspace is the virtual environment created by interconnected networks and digital devices

Key Elements

- Networks (e.g., the internet, intranets)
- Digital devices (e.g., computers, smartphones)
- Data (e.g., personal, financial, governmental information)

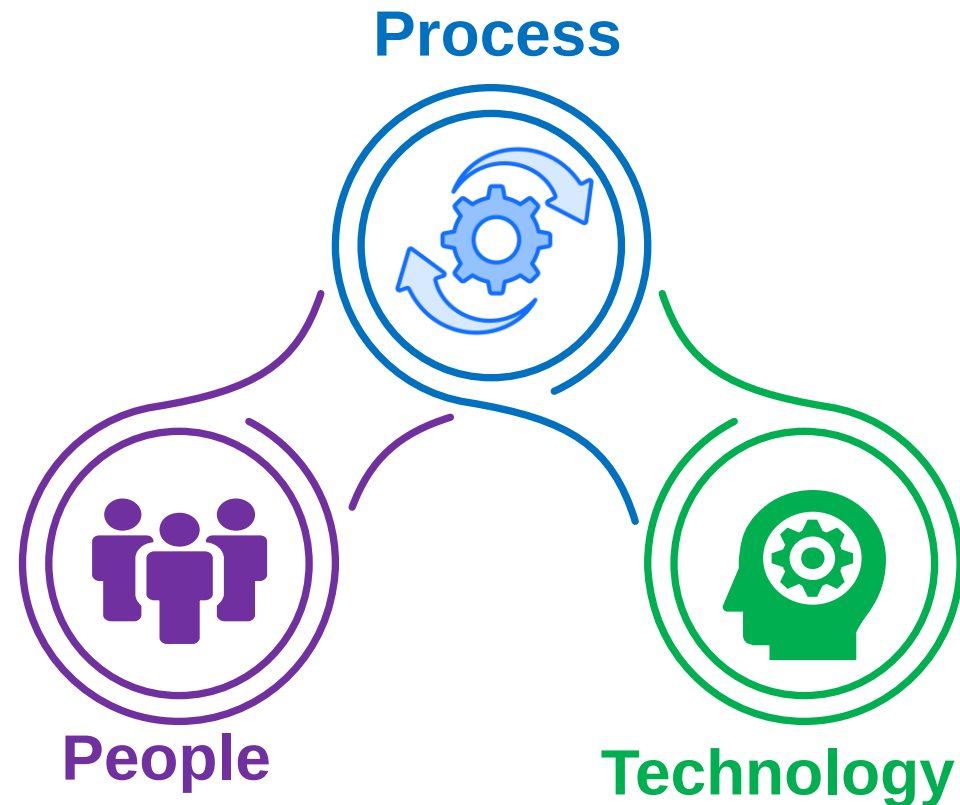


Definition

Cyber Security involves the practices to protect systems, networks, and data from digital threats

Pillars of Cyber Security

1. **People**: Training and awareness of all individual
2. **Process**: Frameworks, policies, and procedures
3. **Technology**: Tools and technologies used





WHY AWARENESS IS IMPORTANT?

Cyber Security Awareness

About people understanding the risks they face online and knowing how to protect themselves

Importance

Cybersecurity is
Collective Responsibility



COMMON CYBER ATTACKS



PHISHING

- **Scam** where someone **pretends to be a trusted** source
- **Steals** personal information
 - Login credentials
 - Financial information
 - Personal or organisational data

PHISHING

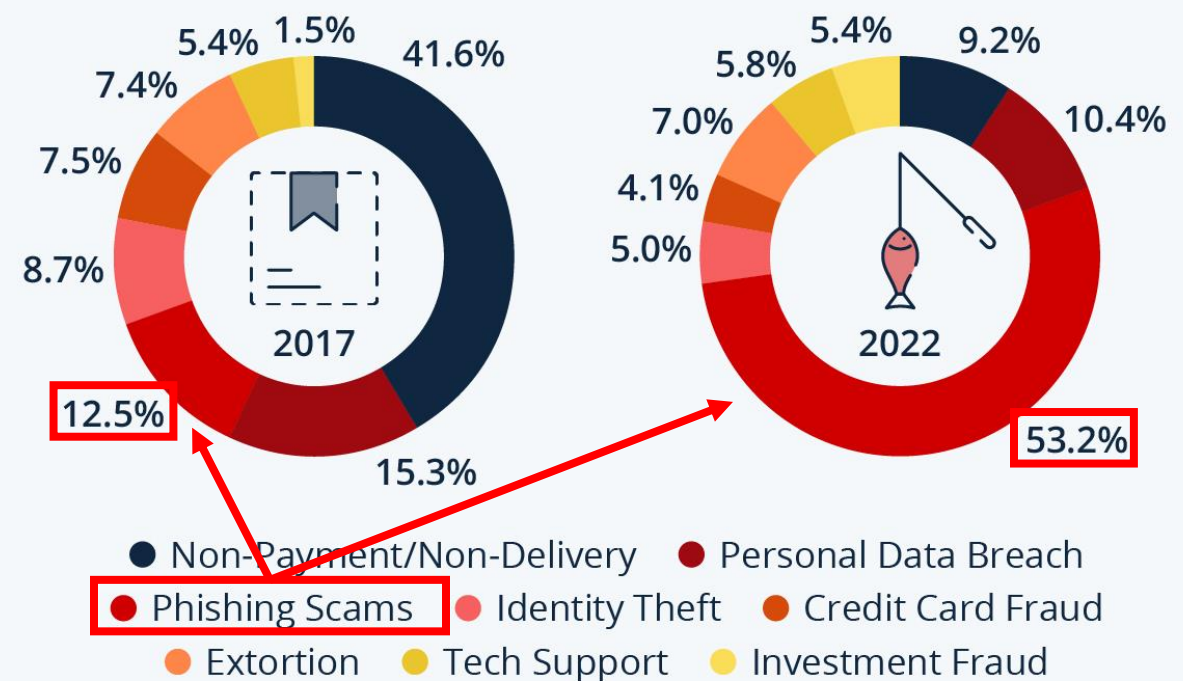
- Roughly **156 million** phishing emails are sent globally every day¹
- Phishing is the root of **95%** of all successful attacks worldwide²
- **92%** phishing attacks are done via email, rest via mobile sms or messaging apps³

Ref:

1. Symantec Security Technology and Response Group
2. <https://cybercrime.doj.gov.ph/what-is-phishing/> (last visited: 14 Sep 24)
3. <https://www.nu.edu/blog/cybersecurity-statistics/> (last visited: 14 Sep 24)

The Most Prevalent Forms of Cyber Crime

Share of worldwide cyber attacks by type



Sources: Statista Market Insights NCSO, FBI, IMF

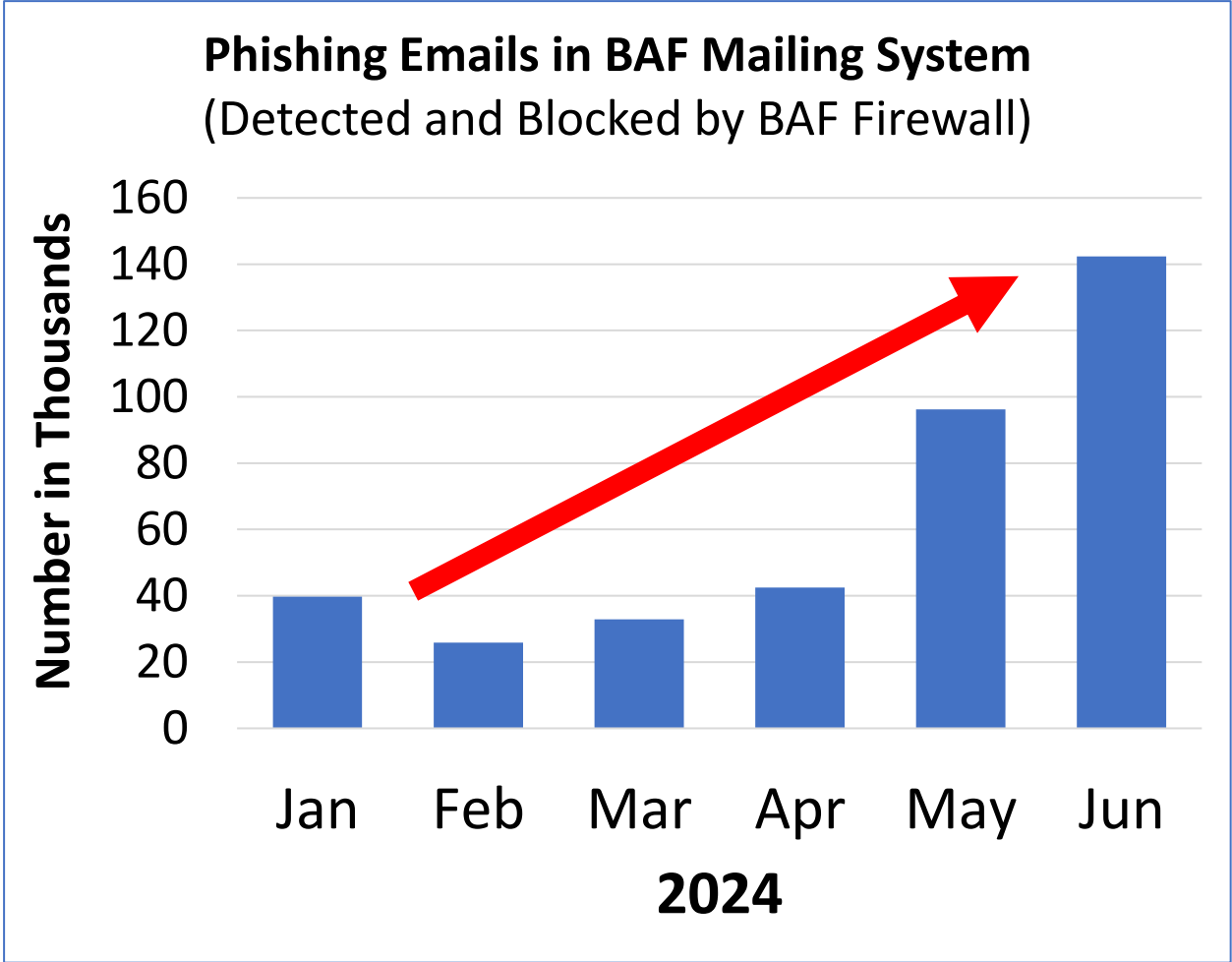


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ANATOMY OF A PHISHING EMAIL



Form: 4suef856seuBaf@drleebrowser.com

04 September, 2024

To: airsecy@baf.mil.bd

Hello Air Secy,

False Sense of Urgency

We received an instruction to delete your account.

We will process the req within 24 Hours.

All features associated with the account will be lost.

To retain your account, Kindly Cancel your request to continue using our services

Cancel Request



<https://www.imperva.com/learn/applicationsecurity/phishing-attackscam/#:~:text=Phiss.sa>

**Actual Link on
Mouse hover**



ANATOMY OF A PHISHING EMAIL

hello2@melecious.email-check.bd

Spoofed Email Address

03 September, 2024

To:

trgwg@baf.mil.bd

Sir, Assalamu Alaikum,

Subject letter attached for your kind action please.

https://drive.google.com/drive/folders/1AljQssqzfGlzM9cjQ_O637nOCkcdD1V0?usp=sharing

URL:<https://api.qrserver.com/v1/ddrnht/?size=300x300&data=https%3A%2F%2Fappurl.io%2FHU0ISnM70y>

**Actual Link on
Mouse hover**



ANATOMY OF A PHISHING EMAIL



Form: Plan and Project Section<staffinfoao@baf.mil.bd>

To: ddsupprog1@baf.mil.bd

 Update on 47 X ...s for C-130 Ac.pdf (6.9 MB) [Download](#) | [Briefcase](#) | [Remove](#)

Assalamu alaikum Sir,

Please see the attached file for your .. info and act pl.

Regards

Plan and Project Section
Ops & Plan Directorate
Armed Forces Division





PHISHING

<https://mail.baf-mil-bd.com/>

<https://mail.baf.mil.bd/>

Bangladesh Air Force
Sign in Help: BAF Ext- 3524 ,3525

Username:

Password:

☐ Stay signed in

Version: [What's This?](#)

CLOINED LOGIN PAGE

Zimbra :: the leader in open source messaging and collaboration :: Blog - Wiki - Forums
Copyright © 2005-2023 Synacor, Inc. All rights reserved. "Zimbra" is a registered trademark of Synacor, Inc.



PHISHING

ORG	ORIGINAL DOMAIN	PHISHING URL/DOMAIN
	mail.afd.gov.bd	mailafd-govbd-signin-true.com
	mail.navy.mil.bd	mail-navy-mil-bd-login.com
	mail.dscsc.mil.bd	mail-dscsc-mil-bd.app
	mail.baf.mil.bd	github.io/mail.baf.mil.bd
	rab.gov.bd	rab-gov-bd.gq

Ref: ICT Division, Bangladesh Cyber Threat Landscape 2022 (e-GOV CIRT)



PHISHING

Example: Cloned Facebook

- <https://fäcebook.com>
- <https://faceb00k.com>
- <https://facebook.co>
- <https://fácebook.com>
- <https://facebok.com>
- <https://facebookk.com>
- <https://faceebook.com>

Example: Cloned BAF Mail

- <https://maíl.baf.mil.bd>
- <https://mall.baf.mil.bd>
- <https://mâil.baf.mil.bd>
- <https://maill.baf.mil.bd>
- <https://mail.baf.mil.bdd>
- <https://mail.baf.mill.bd>



PHISHING

Example: Cloned Facebook

- <https://fäcebook.com>
- <https://faceb00k.com>
- <https://facebook.co>
- <https://fácebook.com>
- <https://facebok.com>
- <https://facebookk.com>
- <https://faceebook.com>

Example: Cloned BAF Mail

- <https://maíl.baf.mil.bd>
- <https://mail.baf.mil.bd>
- <https://mâil.baf.mil.bd>
- <https://mail.baf.mil.bd>
- <https://mail.baf.mil.bdd>
- <https://mail.baf.mil.bd>

Phishing Attack

DTE CW&IT





MALWARE

Any software intended to **damage disable** or give someone **unauthorized** access to your computer or other internet-connected device.

Ransomware is a type of malware that **locks all the files** of a computer using encryption and demands for ransom in exchange of the key to unlock your files.



MALWARE

- Malware attacks rose by **71%** between 2016 and 2021.¹
- At any given time, **4.1 million** sites are infected with malware.²
- On average, a ransomware attack costs a business **\$4.91 million**.³

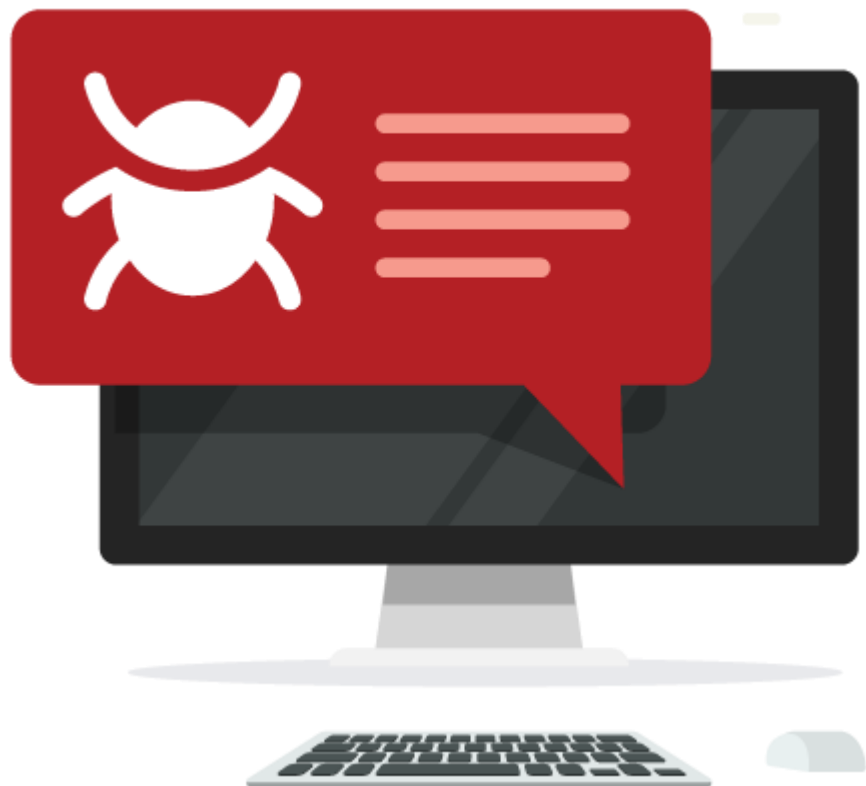
Example

- Viruses/ Worms
- Adware
- Spyware
- Ransomware
- Malicious Browser Extensions/Plugins

Refs:

1. <https://www.sciencedirect.com/science/article/pii/S1319157822002762?via%3Dihub>
2. <https://www.sitelock.com/resources/security-report/>
3. <https://www.ibm.com/reports/data-breach>

MALWARE



Propagation Methods

- Malicious Email Attachments
- Pirated and Cracked Software
- Malicious Browser Extensions
- Infected Removable Media
- Malicious websites



MALWARE IN AN EMAIL



Form: info.e.notification@mod.gov-loan.com

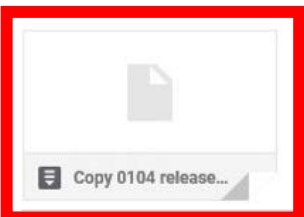
04 September, 2024

To: ddcwit@baf.mil.bd

Assalamualikum,

We are pleased to inform you that your Car Loan for Military Officers has been approved. To finalize your loan and receive the disbursement **Immediate action** is required.

Please download the attached instructions. Kindly follow them strictly.



Best Regards,

Lt. Col. Rahman

Director, Financial Services

Ministry of Defense, Bangladesh



MALWARE: PIRATED AND CRACKED SOFTWARE

Pirated software is unauthorized and illegally copied software that is distributed or sold **without the permission** of the software's owner.

Cracked software is also pirated, but these are **modified** to bypass copy protection.

MALWARE: PIRATED AND CRACKED SOFTWARE

Piracy Methods

- Using common serial numbers
- Using a crack file
- Using activator tool
- Using pre-activated copy
- Using serial numbers bought from unauthorized sources



ট150

Genuine Products tools



ট399

Microsoft Office 365 Ac...

If one year license of Office 365 is **BDT 9000** (approx),
how can someone sell it in **BDT 300** only?

MALWARE: MALICIOUS BROWSER EXTENSIONS

Browser Extension/Plugin

A software module for customizing a web browser.



Malicious Browser Extension/Plugin

- Steal passwords and other personal data
- Access data on computers
- Display unwanted ads



SOCIAL ENGINEERING

When someone tricks you into revealing sensitive information by **exploiting** your trust and hacks into your accounts **without actively attacking** the digital device/system.



SOCIAL ENGINEERING: CASE STUDY

A 'friend' in Facebook posts '**Four importance things about me**' and encourages you to do the same

- Favourite Pet Name
- Place of Birth
- Favorite Movie
- Your First School

These information can be related to:

- **Answer to secret questions** for recovering any online account
- **Choice of password**

By responding to a fun-to-share act on social media,
you are compromising your security!



REAL-LIFE EXAMPLES



LEAKED PERSONAL INFORMATION

Username & Password of Personal Accounts

SOFT: Chrome

URL: <https://www.daraz.com.bd/contact-us/>

USER: [cadet\[REDACTED\].23@gmail.com](mailto:cadet[REDACTED].23@gmail.com)

PASS: 7nSkY[REDACTED]An5E

SOFT: Chrome

URL: <https://m.facebook.com/>

USER: [fp12\[REDACTED\]@gmail.com](mailto:fp12[REDACTED]@gmail.com)

PASS: xpwin9[REDACTED]520@cadet

SOFT: Chrome

URL: <https://mail.baf.mil.bd/>

USER: fahim[REDACTED]1

PASS: Fahim[REDACTED]023

SOFT: Edge

URL: Trust App

USER: TBL[REDACTED]447

PASS: Songi[REDACTED]823



LEAKED OFFICIAL DOCUMENT



BAF Computer Screenshot Taken by Hacker

CONFIDENTIAL

1st Witness : BD/4692 [REDACTED] Khandaker, Log Asst., 5 Sqn BAF states:

"I have been serving as a Log Asst of 5 Sqn BAF since 20 Feb 22. The colour printer (Brand: Epson, Model: L-130, Ser No: 8885007024493) was issued to 5 Sqn BAF on 11 Dec 2001, vide ref: Air HQ/10136/Air Trg/V-71/22A dt: 22 Jul 2001 as per log book. This colour Printer has become unsvc on 29 Mar 21. I do here by produce the BAF F-674 & F-675 log book and photograph of unsvc colour printer as Exhibit 'A', 'B' & 'C'."

Exhibit- 'A', 'B' & 'C'

Can you find the
the mistake?

1 2 3 4 5 6 7 8 9



Cyber Security Awareness Measures



MYTHS VS REALITY

- Cybersecurity is **too complex**. In reality, it is **very simple**
- Cyber attacks are **sophisticated** and they are unstoppable. However, they are mostly **non-sophisticated**
- Cyber attacks are **directed toward specific targets**. Practically, anyone can be a target. The **weakest link** of cyber security is often **people**

60% cyber security **risks can be reduced through awareness** programs¹

Ref:

1. <https://www.forbes.com/councils/forbestechcouncil/2023/10/24/how-cybersecurity-training-lowers-risk-among-employees/>



CYBER SECURITY AWARENESS MEASURES

CYBER SECURITY IS AS SIMPLE AS **A B C**

A

ALWAYS

B

BE

C

CAREFUL

CYBER SECURITY AWARENESS MEASURES

DO NOT TAKE THE BAIT



DO

- ✓ Be cautious
- ✓ Verify the sender
- ✓ Check for spelling and grammar errors
- ✓ Hover the cursor before you click
- ✓ Keep software up to date

CYBER SECURITY AWARENESS MEASURES

DO NOT TAKE THE BAIT



DON'T

- ✗ Don't Click on suspicious links
- ✗ Don't Download attachments from unknown senders
- ✗ Don't Provide personal information
- ✗ Don't Trust urgent or threatening messages
- ✗ Don't Click on any email or social media messages you don't expect or need

CYBER SECURITY AWARENESS MEASURES

**TREAT YOUR
PASSWORD
LIKE YOUR
TOOTHBRUSH**



- Do not share your passwords
- Change your password every three months
- Pick a good (strong) one



CYBER SECURITY AWARENESS MEASURES

USE **UNIQUE** & **STRONG** PASSWORD

- At least **12 characters** long
- Contains:
 - ✓ a sequence of **upper & lower-case letters, numbers and symbols**, or
 - ✓ a memorable **phrase of 4 – 7 unrelated words**
- **Unique and different** password for each account
- **Difficult to guess** without using one's name, service number, phone number, address, pet's name, school or favourite team, etc



CYBER SECURITY AWARENESS MEASURES

EXAMPLES OF **UNIQUE** & **STRONG** PASSWORD

- Utilize Phrases or Sentences: **HotTeaOn1RainyDay!**
- Use Password Padding: **^(Coffee4Dinner)^**
- Use Mixed Letter with substitution: **MdR@k1bulH0qu3%1oo2E**
MdRakibulHoque%10023

Strong Passwords:

- ☐ At least **12 characters** long
- ☐ Contains:
 - ✓ a sequence of **upper & lower-case letters, numbers and symbols**, or
 - ✓ a memorable **phrase of 4 – 7 unrelated words**

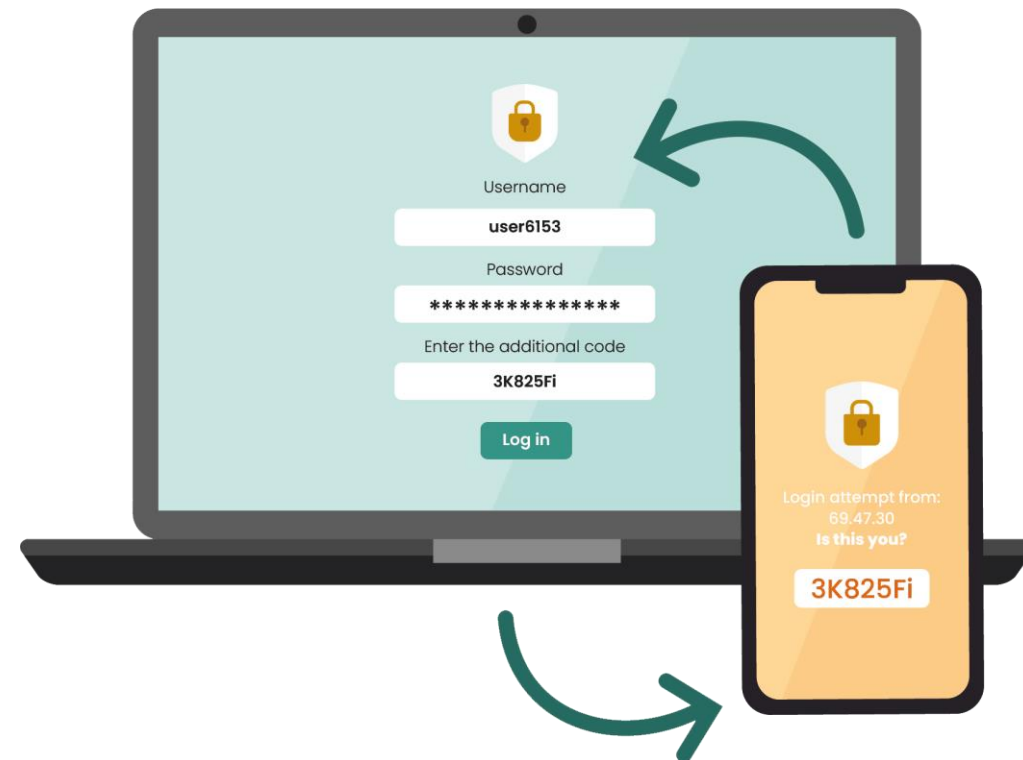
Substitution

- \$, S or 5 for s
- 1, l or ! for i
- @ or A for a
- 7 or T for t
- 3 or E for e
- 9, G or 6 for g
- 0 or O for o
- 8 or B for b

CYBER SECURITY AWARENESS MEASURES

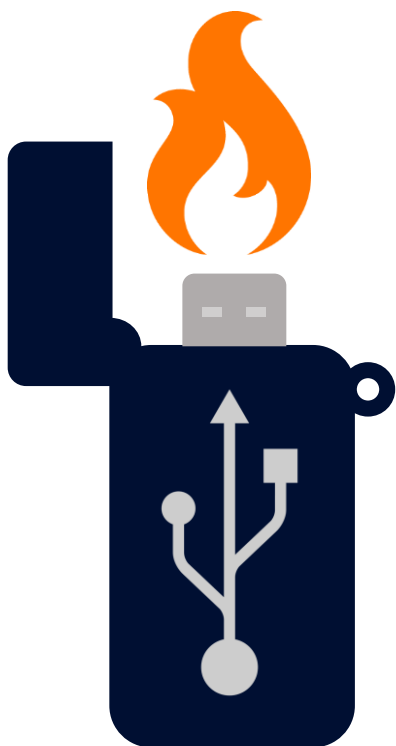
USE MULTI-FACTOR AUTHENTICATION

- Two-Step Verification
- Second verification method:
 - OTP sent to your mobile
 - OTP sent to or email
 - Authenticator app or
 - Mobile app confirmation

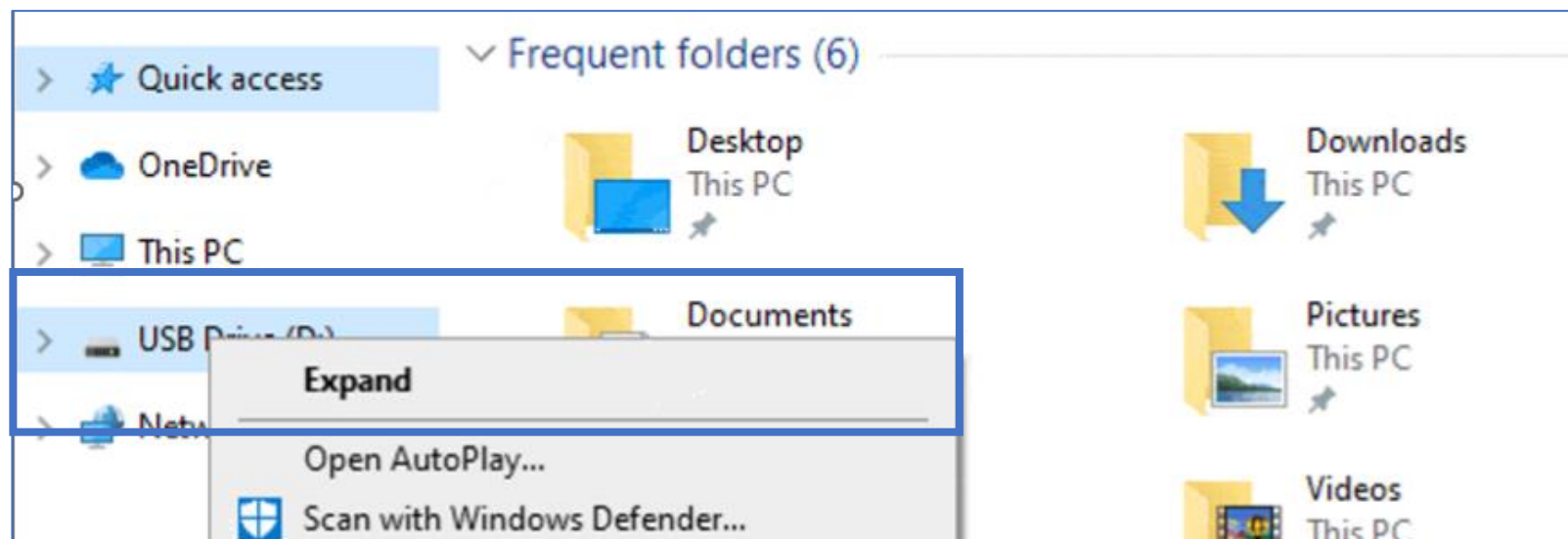


CYBER SECURITY AWARENESS MEASURES

REDUCE USE OF FLASH DRIVE



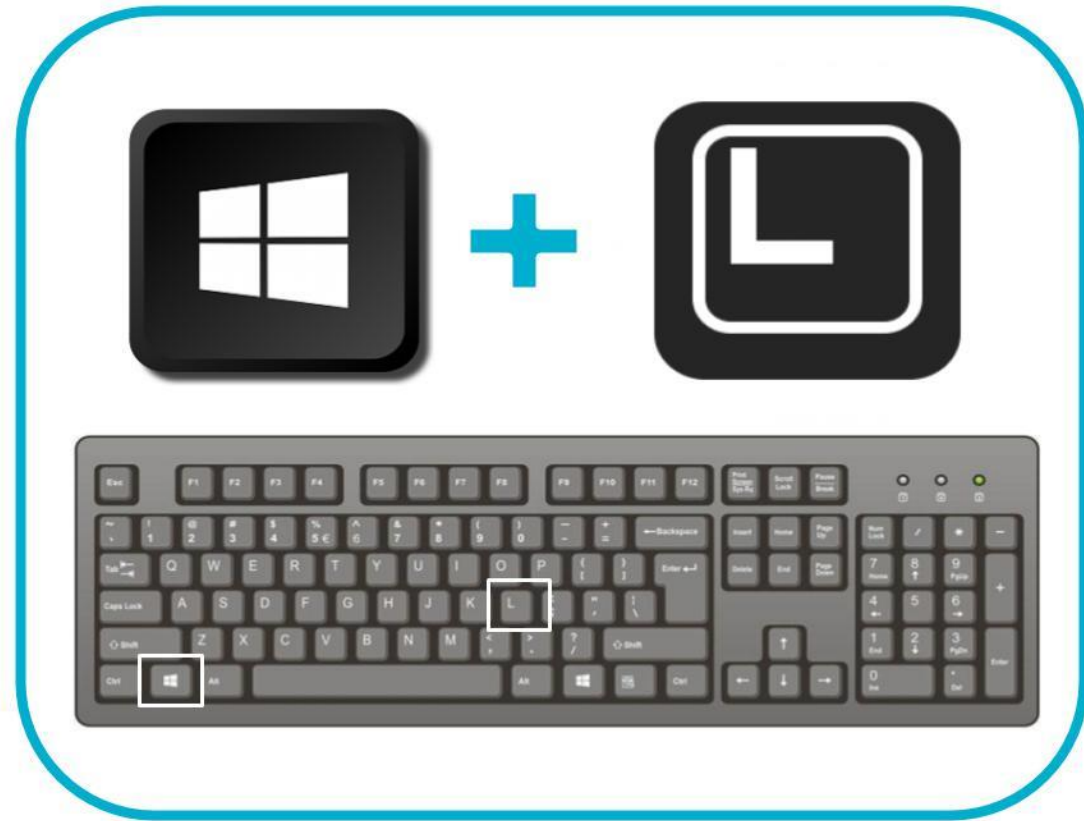
- Do not use flash drives except unavoidable situations
- Use antivirus to check before accessing
- Do not double click, always use explorer to access



CYBER SECURITY AWARENESS MEASURES

LOCK YOUR COMPUTER

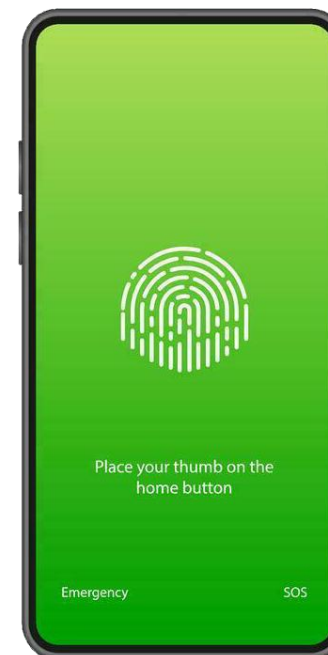
Keep your data secured by locking your computer and devices when you are not at your desk.



CYBER SECURITY AWARENESS MEASURES

Keep Your Home Screen Locked

- Passcode
- Pattern
- Fingerprint
- Face recognition





CYBER SECURITY AWARENESS MEASURES

STOP USING PIRATED AND CRACKED SOFTWARE
USE FREE OPENSOURCE SOFTWARE

Do we want to use pirated software
and take the risks of cyber attack?

or

Do we want to use free opensource software,
which may take some time to learn?



CYBER SECURITY AWARENESS MEASURES



MOBILE DEVICE SECURITY

- Keep **Bluetooth Turned Off**
- Keep Device's Software **Up-to-Date**
- Download Apps From **Trusted Sources**
- Only Connect to **Secure Wi-Fi**



GET IT ON

Google Play



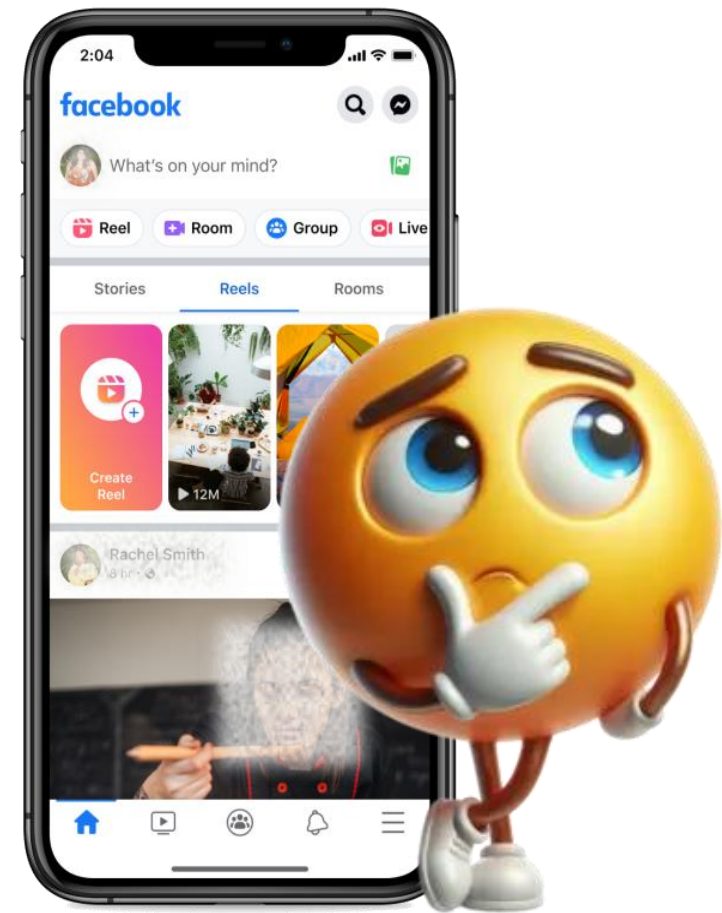
Download on the
App Store



CYBER SECURITY AWARENESS MEASURES

**THINK BEFORE SHARING
YOUR PRIVATE LIFE
ON SOCIAL MEDIA**

Posting too much of your private life on social media can invite unwanted issues which you could never anticipate.





CYBER SECURITY AWARENESS MEASURES



DO NOT SHARE CLASSIFIED INFORMATION

Do not share official information online in any public or private group or platform without authorization





CYBER SECURITY AWARENESS MONTH

01-31 October 2024



CYBER SECURITY AWARENESS FOR A SAFER TOMORROW

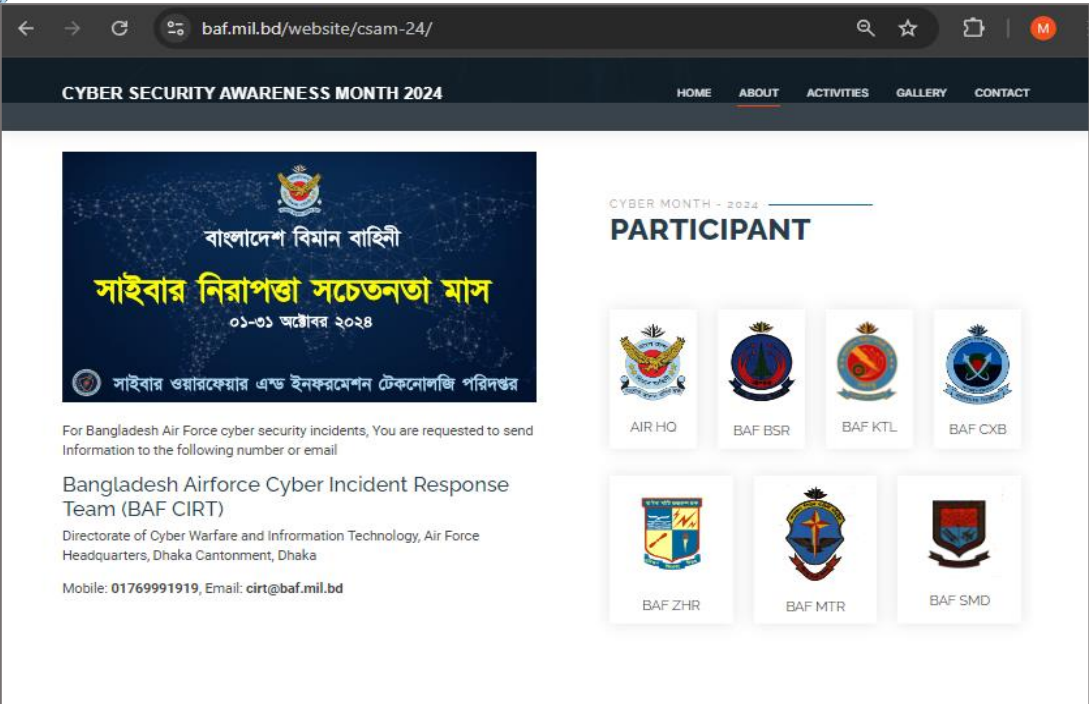
**DIRECTORATE OF CYBER WARFARE AND INFORMATION TECHNOLOGY
BANGLADESH AIR FORCE**



Summary



Q&A: CONTACT



<https://www.baf.mil.bd/website/csam-24/>



Scan to Download





REPORTING CYBER INCIDENTS

BAF CIRT

Mobile: +8801769991919

Email: cirt@baf.mil.bd



Scan To Report



Thank You